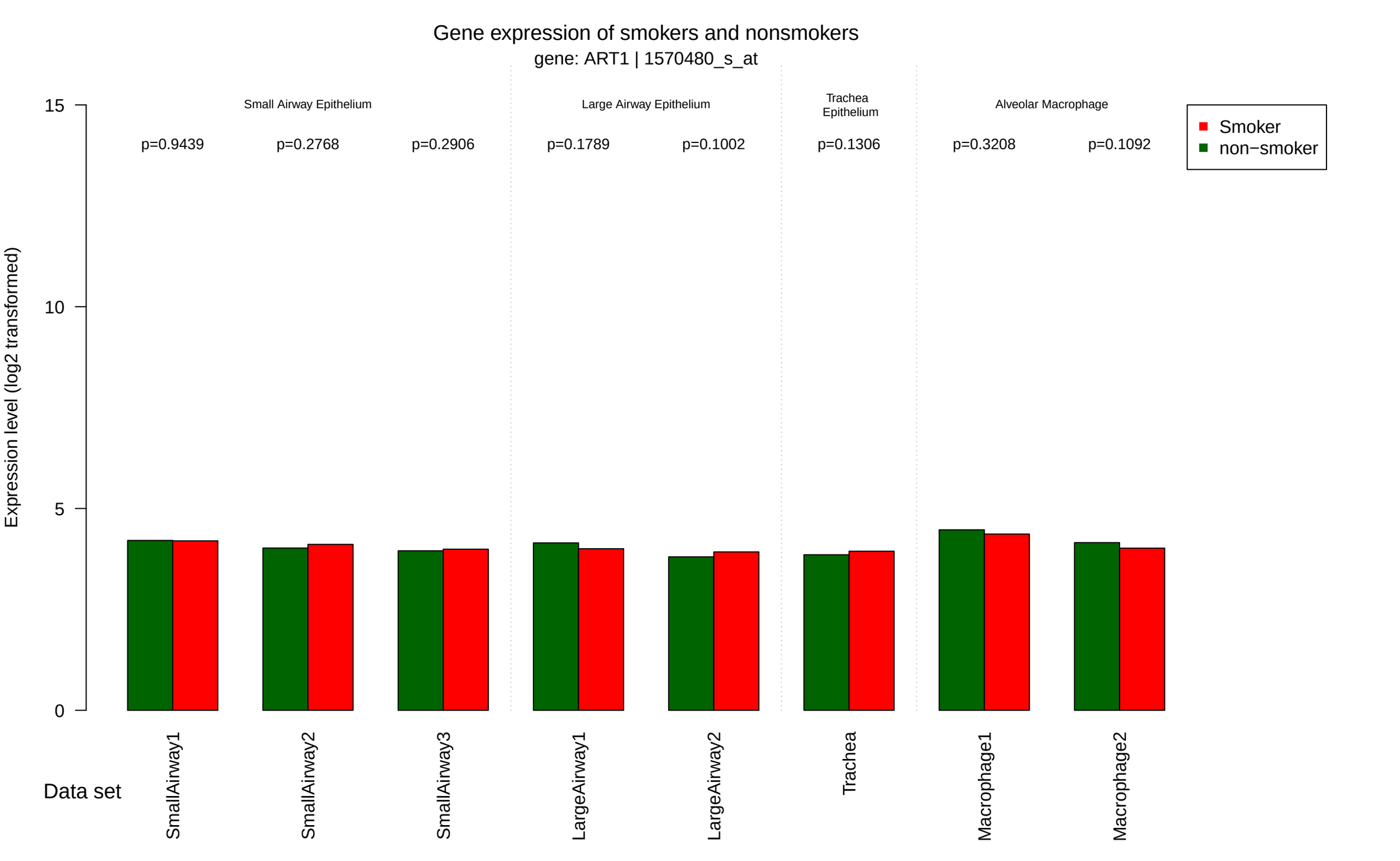
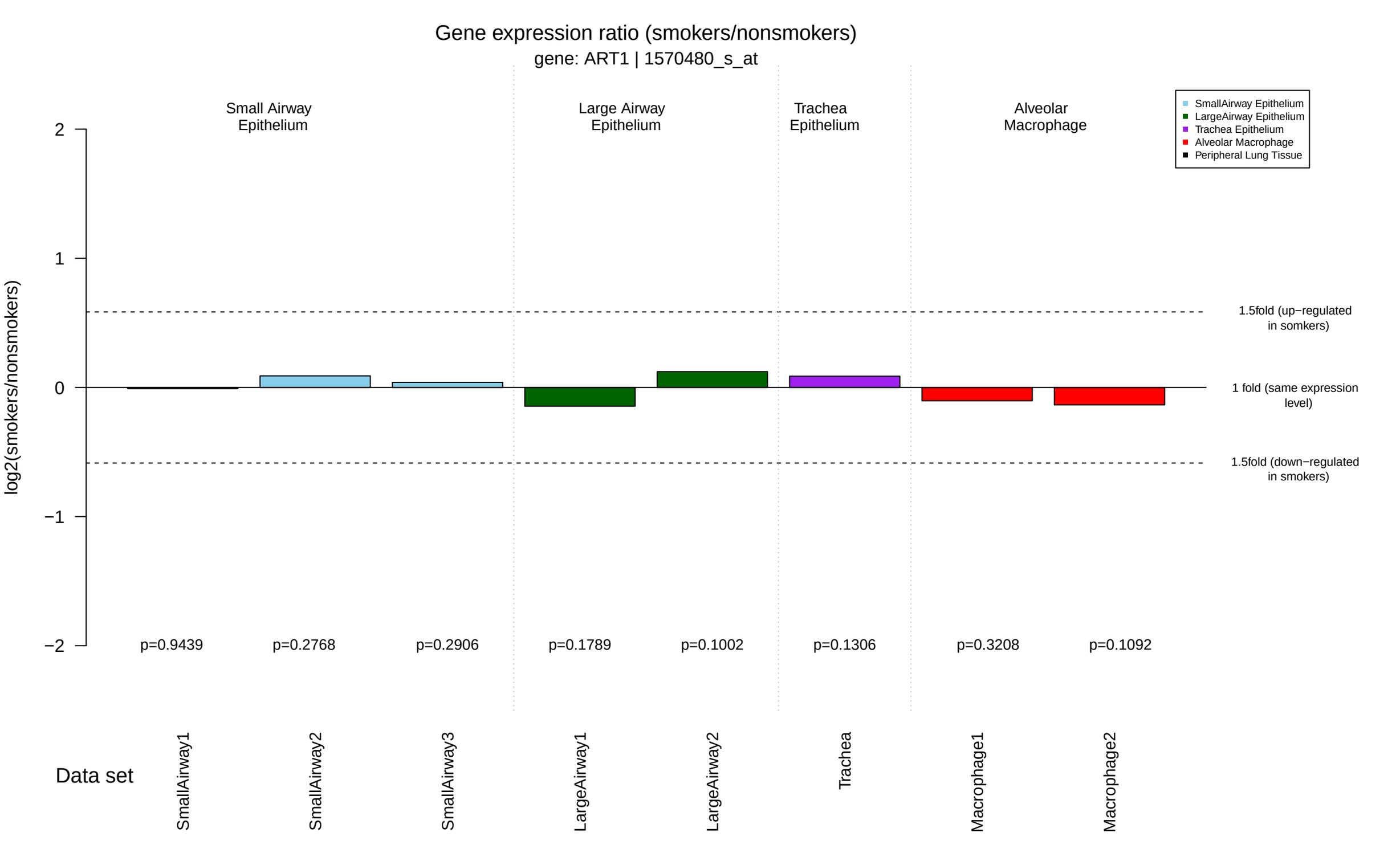
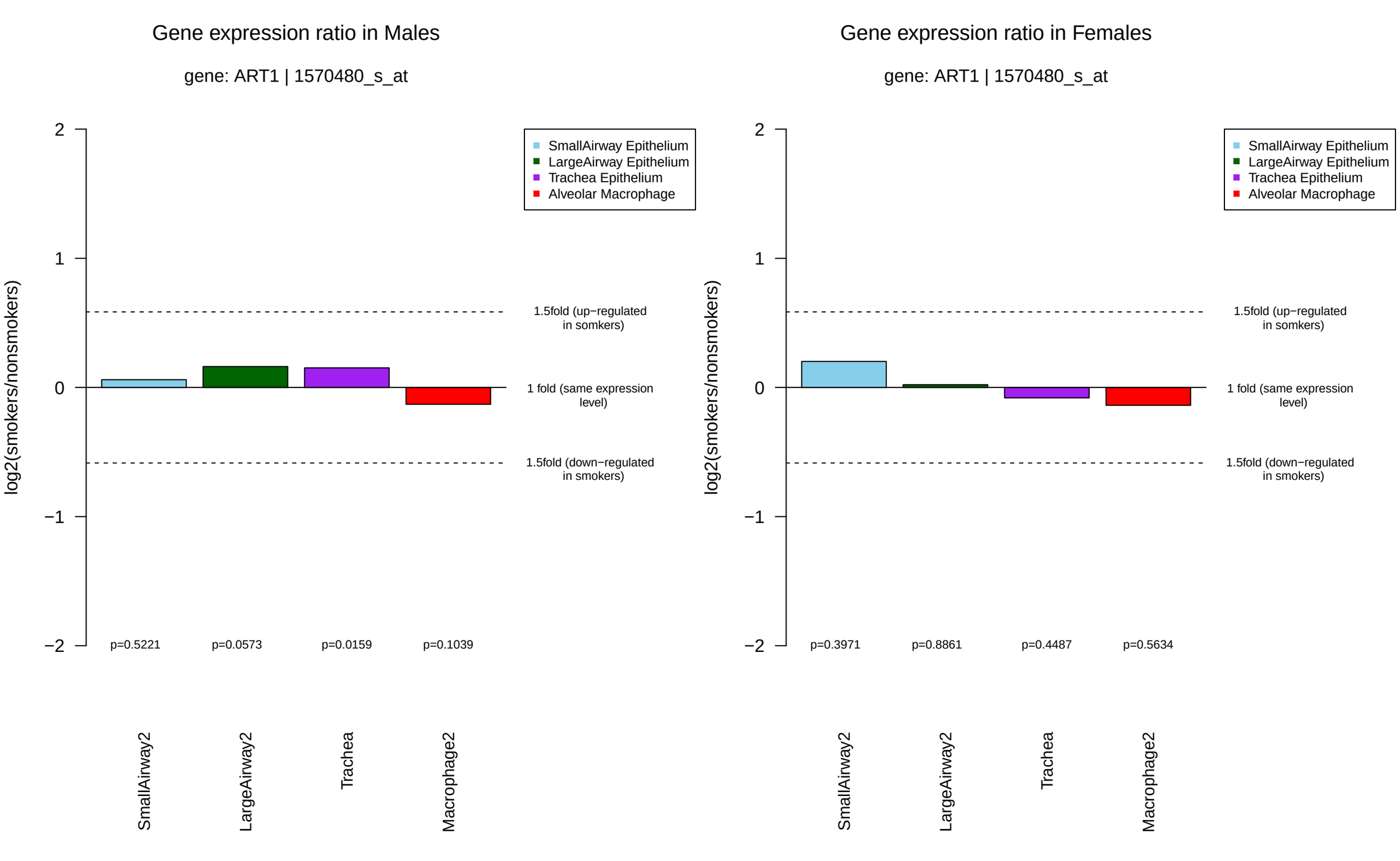
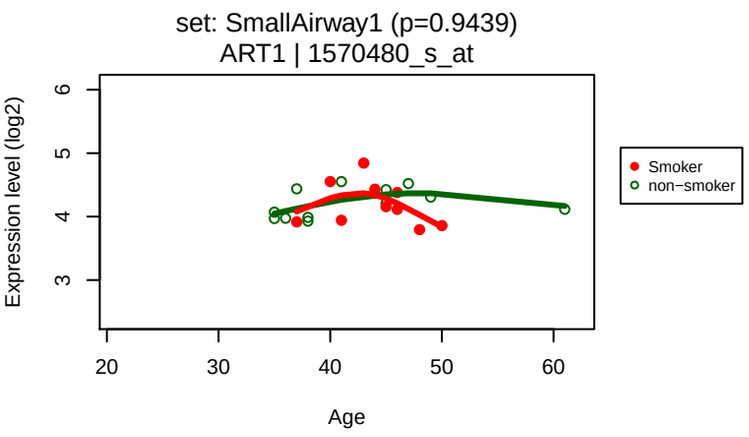
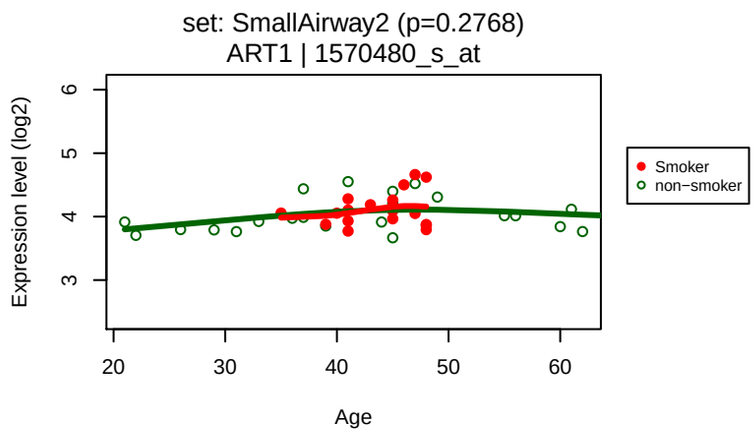
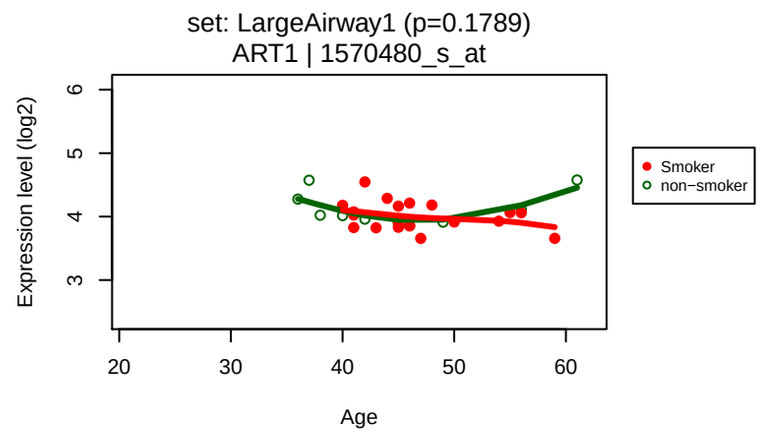

Gene expression of smokers and nonsmokers
gene: ART1 | 1570480_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9439
p=0.2768
p=0.2906
p=0.1789
p=0.1002
p=0.1306
p=0.3208
p=0.1092
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

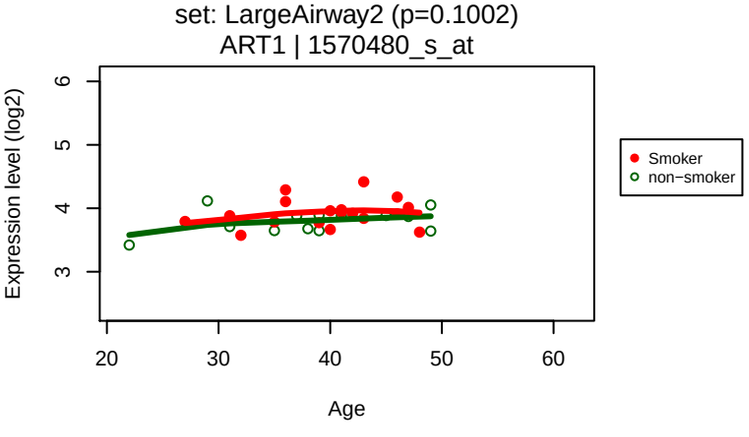

Gene expression ratio (smokers/nonsmokers)
gene: ART1 | 1570480_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9439
p=0.2768
p=0.2906
p=0.1789
p=0.1002
p=0.1306
p=0.3208
p=0.1092
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

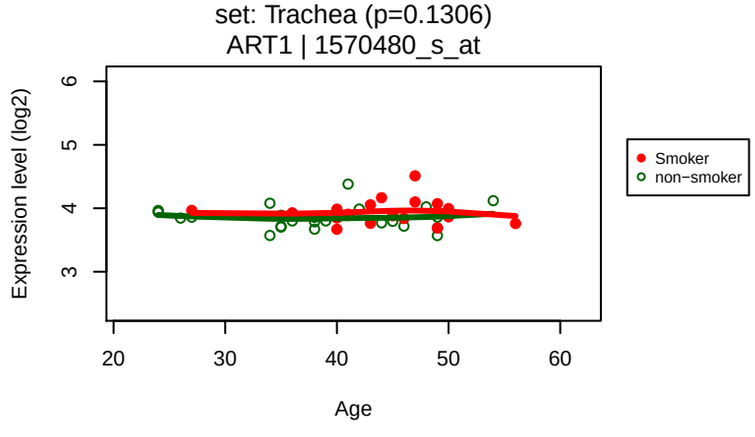

Gene expression ratio in Males
gene: ART1 | 1570480_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5221
p=0.0573
p=0.0159
p=0.1039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ART1 | 1570480_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3971
p=0.8861
p=0.4487
p=0.5634
SmallAirway2
LargeAirway2
Trachea
Macrophage2

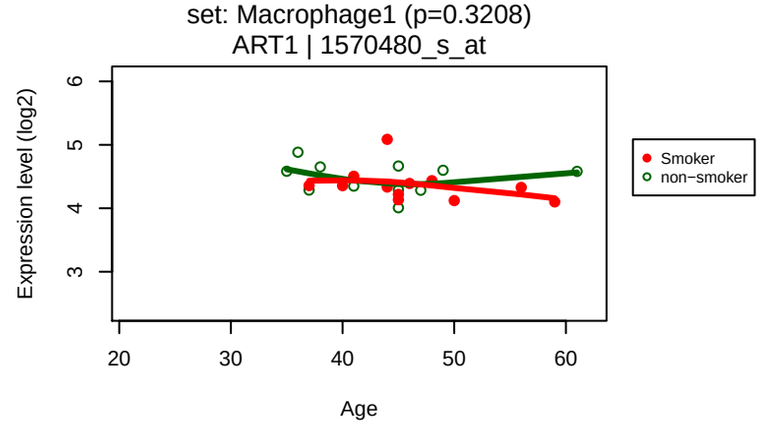

set: SmallAirway1 (p=0.9439)
ART1 | 1570480_s_at
Expression level (log2)
Age
Smoker
non−smoker

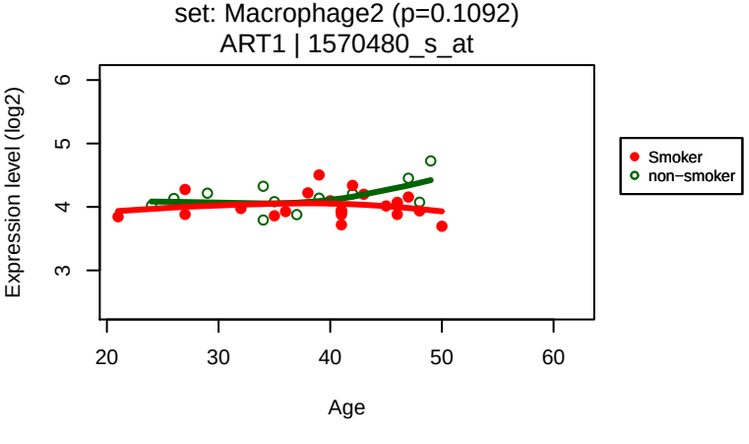

set: SmallAirway2 (p=0.2768)
ART1 | 1570480_s_at
Expression level (log2)
Age
Smoker
non−smoker

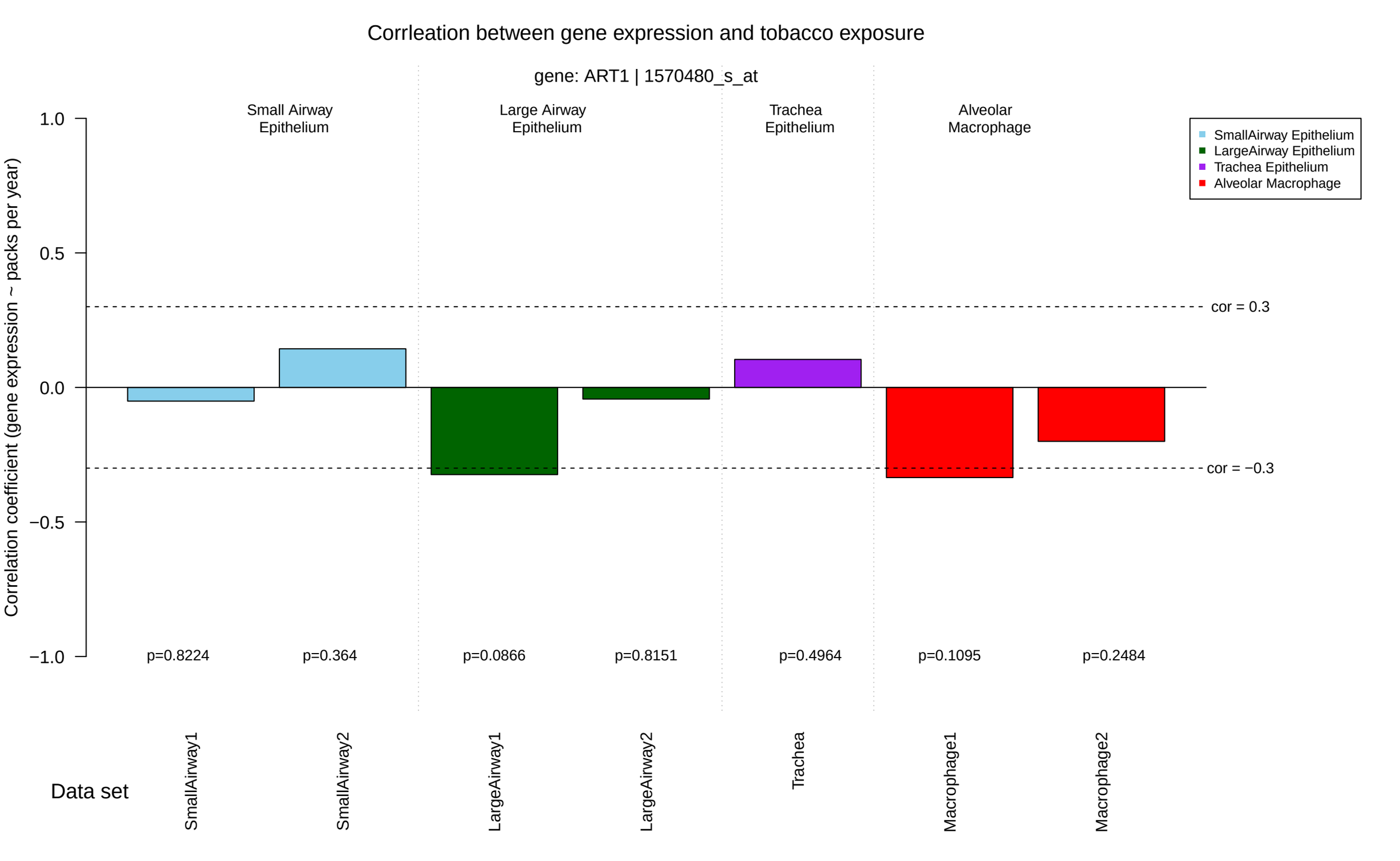

set: LargeAirway1 (p=0.1789)
ART1 | 1570480_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1002)
ART1 | 1570480_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1306)
ART1 | 1570480_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3208)
ART1 | 1570480_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1092)
ART1 | 1570480_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ART1 | 1570480_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8224
p=0.364
p=0.0866
p=0.8151
p=0.4964
p=0.1095
p=0.2484
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2